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JupyterLab Python [conda env:base] Anaconda Toolbox

Name ↑ Actions

▼ DataScience

DataScience.ipynb

```
[5]: #Regression
#Gradient Descent
import numpy as np
#Dataset
X=np.array([1,2,3,4])
y=np.array([6,7,8,9])
#Initial values
m=0 #Start with random values
b=0
learning_rate=0.01
epochs=1000
n=len(X)
for i in range(epochs):
    #predictions
    y_pred=m*X+b #current line equation
    #Gradients
    dm=(-2/n) *np.sum(X*(y-y_pred)) # how much to change slope
    db=(-2/n) *np.sum(y-y_pred) #how much to change intercept
    #Update parameters
    m=m-learning_rate*dm
    b=b-learning_rate*db #move toward Lower error
print("Slope(m):",m)
print("Intercept(b):",b)

Slope(m): 1.070840680776193
Intercept(b): 4.791719759731761

[13]: #OLS(Ordinary Least Square)
import pandas as pd
from sklearn.linear_model import LinearRegression
#Sample Data
```

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Python [conda env:base] v4.20.0

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Code

print("Intercept(b):",b)

Slope(m): 1.070840680776193

Intercept(b): 4.791719759731761

[13]:

```
#OLS(Ordinary Least Square)
import pandas as pd
from sklearn.linear_model import LinearRegression
#Sample Data
df=pd.DataFrame({
    'Experience':[1,2,3,4,5],
    'Salary':[30000,35000,40000,45000,50000]
})
#Input Feature
X=df[['Experience']]
#Target variable
y=df['Salary']
#Create Model
model=LinearRegression()
#Train Model(OLS happens here)
model.fit(X,y)
#Predict salary
predictions=model.predict(X)
#Results
print("Slope:",model.coef_[0])
print("Intercept:",model.intercept_)
print("Predicted Salaries:",predictions)
new_experience=pd.DataFrame({'Experience':[6,7,8,9]
})
#Predict salary fro new values
new_predictions=model.predict(new_experience)
print("Predicted Salaries for 6,7,8,9:")
print(new_predictions)
```

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```
print("Predicted Salaries:", predictions)
new_experience=pd.DataFrame({'Experience':[6,7,8,9]
})
#Predict salary fro new values
new_predictions=model.predict(new_experience)
print("Predicted Salaries for 6,7,8,9:")
print(new_predictions)
```

Slope: 5000.000000000001
Intercept: 24999.999999999996
Predicted Salaries: [30000. 35000. 40000. 45000. 50000.]
Predicted Salaries for 6,7,8,9:
[55000. 60000. 65000. 70000.]

[14]: `import matplotlib.pyplot as plt`

`plt.figure(figsize=(8,5))`

`# Original data points`

`plt.scatter(df['Experience'], df['Salary'], color='blue', label='Actual Data')`

`# Regression Line (trained data)`

`plt.plot(df['Experience'], predictions, color='red', label='Regression Line')`

`# New predictions`

`plt.scatter(new_experience['Experience'], new_predictions, color='green')`

`# Labels`

`plt.xlabel("Experience (Years)")`

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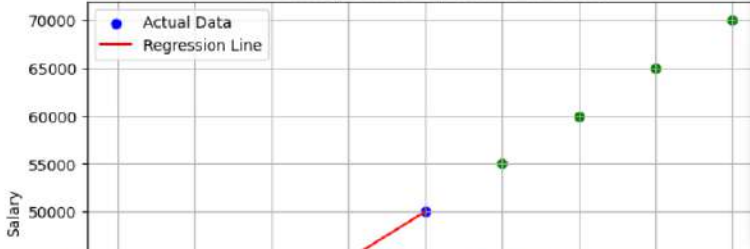
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```
# New predictions
plt.scatter(new_experience['Experience'], new_predictions, color='green')

# Labels
plt.xlabel("Experience (Years)")
plt.ylabel("Salary")
plt.title("Linear Regression: Experience vs Salary")
plt.legend()
plt.grid()
plt.show()
```

Linear Regression: Experience vs Salary



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30000

1 2 3 4 5 6 7 8 9

Experience (Years)

```
[17]: import pandas as pd
from sklearn.linear_model import LinearRegression
df=pd.DataFrame([
    'Area':[1000,1200,1500,1800,2000],
    'Bedrooms':[2,3,4,5,6],
    'Age':[10,5,8,4,3],
    'Price':[30000,35000,45000,50000,60000]
])
X=df[['Area','Bedrooms','Age']]
#Target variable
y=df['Price']
#Create Model
model=LinearRegression()
#Train Model(OLS happens here)
model.fit(X,y)
#Predict salary
predictions=model.predict(X)
#RESULTS
predictions=model.predict(X)
#RESULTS
print("Slope:",model.coef_)
print("Intercept:",model.intercept_)
print("Predicted Price:",predictions)
new=pd.DataFrame({'Area':[12000,15000,16000,17000,18000],
                  'Bedrooms':[6,7,8,9,5],
                  'Age':[11,12,13,14,15]
})
#Predict salary for new values
```

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Anaconda Toolbox v4.20.0

Name ↑ Actions

DataScience

DataScience.ipynb

```
print("Slope:",model.coef_)
print("Intercept:",model.intercept_)
print("Predicted Price:",predictions)
new=pd.DataFrame({'Area':[12000,15000,16000,17000,18000],
                  'Bedrooms':[6,7,8,9,5],
                  'Age':[11,12,13,14,15]
})
#Predict salary fro new values
new_predictions=model.predict(new)
print("Predicted Price for :")
print(new_predictions)

Slope: [ -20.83333333 14166.66666667  833.33333333]
Intercept: 13583.3333333327
Predicted Price: [29416.66666667 35250.          45666.66666667 50250.
 59416.66666667]
Predicted Price for :
[-142250.        -189750.        -195583.33333333 -201416.66666667
 -278083.33333333]
```

[18]: #VIF Code

```
import pandas as pd

from statsmodels.stats.outliers_influence import variance_inflation_factor

# Features
X = df[['Area','Bedrooms','Age']]

# VIF Calculation
vif = pd.DataFrame()
```

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The screenshot displays a JupyterLab environment with a Python kernel. The code in the notebook calculates the Variance Inflation Factor (VIF) for a dataset with three features: Area, Bedrooms, and Age. The output shows that the VIF values are 997.295198 for Area, 781.668866 for Bedrooms, and 23.765709 for Age. A Snipping Tool window is open in the bottom right corner, showing a screenshot of the code and output, with a message indicating that the screenshot has been copied to the clipboard and saved to the screenshots folder.

```
[18]: #VIF Code

import pandas as pd

from statsmodels.stats.outliers_influence import variance_inflation_factor

# Features

X = df[['Area', 'Bedrooms', 'Age']]

# VIF Calculation

vif = pd.DataFrame()

vif['Feature'] = X.columns

vif['VIF'] = [

    variance_inflation_factor(X.values, i)

    for i in range(X.shape[1])

]

print(vif)
```

	Feature	VIF
0	Area	997.295198
1	Bedrooms	781.668866
2	Age	23.765709

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JupyterLab Python [conda env:base] Anaconda Toolbox

```
[20]: import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

from sklearn.preprocessing import PolynomialFeatures

from sklearn.linear_model import LinearRegression

# Sample dataset

df = pd.DataFrame([

    'Hours': [1, 2, 3, 4, 5],

    'Marks': [10, 25, 45, 70, 100]

])

X = df[['Hours']]

y = df['Marks']

# Polynomial transformation

poly = PolynomialFeatures(degree=400)

X_poly = poly.fit_transform(X)

# Model training

model = LinearRegression()
```

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The screenshot displays a Jupyter Notebook environment with the following code in the cell:

```
poly = PolynomialFeatures(degree=400)

X_poly = poly.fit_transform(X)

# Model training

model = LinearRegression()

model.fit(X_poly, y)

# Smooth curve generation (IMPORTANT STEP)

X_range = np.linspace(X.min(), X.max(), 100).reshape(-1, 1)

X_range = pd.DataFrame(X_range, columns=['Hours'])

X_range_poly = poly.transform(X_range)

# Predictions for curve

y_curve = model.predict(X_range_poly)

# Plotting

plt.scatter(X, y, color='red', label='Actual Data')

plt.plot(X_range, y_curve, color='blue', label='Polynomial Curve')

plt.title('Polynomial Regression Curve')

plt.xlabel('Hours')
```

A 'Snipping Tool' window is open in the bottom right corner, displaying the message: 'Screenshot copied to clipboard. Automatically saved to screenshots folder.' and a button labeled 'Markup and share'.

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Python [conda env:base] v

Code

```
plt.xlabel('Hours')
plt.ylabel('Marks')
plt.legend()
plt.show()
```

Polynomial Regression Curve

Hours	Marks
1.0	10
2.0	25
3.0	45
4.0	70
5.0	100

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Code

JupyterLab Python [conda env:base] Anaconda Toolbox

1.0 1.5 2.0 2.5 3.0 3.5 4.0 4.5 5.0

Hours

```
[22]: import numpy as np

import matplotlib.pyplot as plt

from sklearn.preprocessing import PolynomialFeatures
from sklearn.linear_model import LinearRegression

X = np.array([1,2,3,4,5,6,7,8,9,10]).reshape(-1,1)
y = np.array([2,5,8,15,28,35,50,68,85,95])

# Degree 1

poly1 = PolynomialFeatures(degree=1)

X1 = poly1.fit_transform(X)

# Degree 2

poly4 = PolynomialFeatures(degree=4)

X4 = poly4.fit_transform(X)

# Degree 9

poly9 = PolynomialFeatures(degree=9)
```

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```
poly4 = PolynomialFeatures(degree=4)

X4 = poly4.fit_transform(X)

# Degree 9
poly9 = PolynomialFeatures(degree=9)

X9 = poly9.fit_transform(X)

model1 = LinearRegression().fit(X1,y)
model2 = LinearRegression().fit(X4,y)
model9 = LinearRegression().fit(X9,y)

plt.scatter(X,y)

plt.plot(X, model1.predict(X1), label="Degree 1")
plt.plot(X, model2.predict(X4), label="Degree 4")
plt.plot(X, model9.predict(X9), label="Degree 9")

plt.legend()

plt.show()
```



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JupyterLab Python [conda env:base] Anaconda Toolbox

```
[ ]: #Degree 1=underfitting -straight line misses the pattern
#Degree 4=good flr-captures the overall curve
#Degree 9=overfitting-tries to pass through every point Learns noise instead of pattern

•[23]: ##Ridge Code

from sklearn.linear_model import Ridge

from sklearn.datasets import make_regression

X, y = make_regression(

    n_samples=100,

    n_features=10,

    noise=20,

    random_state=42

)

ridge = Ridge(alpha=1.0)

ridge.fit(X, y)
```

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```
random_state=42

)

ridge = Ridge(alpha=1.0)

ridge.fit(X, y)

print("Coefficients:")

print(ridge.coef_)

Coefficients:
[62.4844981  98.07649112  57.01070157  52.31272567  35.35343002]

[25]: #Lasso
from sklearn.linear_model import Lasso

from sklearn.datasets import make_regression

X, y = make_regression(

    n_samples=100,

    n_features=10,

    noise=20,

    random_state=42

)

lasso = Lasso(alpha=2.0)
```

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```
noise=20,
random_state=42
)
lasso = Lasso(alpha=2.0)
lasso.fit(X, y)
print("Coefficients:")
print(lasso.coef_)

Coefficients:
[17.88753444 54.76076704  0.          60.9632585  90.0800089  68.30461063
 81.35787563  4.03519677  2.86610797  67.99906953]
```

[20]: #ElasticNet

```
from sklearn.linear_model import ElasticNet
from sklearn.datasets import make_regression

X, y = make_regression(
    n_samples=100,
    n_features=5,
    noise=20,
    random_state=42
```

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26.14321862 36.10757704 13.53982521 20.15695537 16.41401164

```
[28]: #ElasticNet
from sklearn.linear_model import ElasticNet

from sklearn.datasets import make_regression

X, y = make_regression(
    n_samples=100,
    n_features=5,
    noise=20,
    random_state=42
)

elasticnet = ElasticNet(
    alpha=3.0,
    l1_ratio=0.5 # (0.0-1.0)
)

elasticnet.fit(X, y)

print("Coefficients:")
print(elasticnet.coef_)

Coefficients:
[26.14321862 36.10757704 13.53982521 20.15695537 16.41401164]
```

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